MILITARY SPECIFICATION

TRANSMITTER, INDUCTION COMPASS T-611( )/ASN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of Flux Gate designated as Transmitter, Induction Compass T-611( )/ASN (See 6.1 and 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-585	Box, Wood, Wirebound
PPP-B-601	Box, Wood, Cleated Plywood
PPP-B-621	Box, Wood, Nailed & Lock Corner
PPP-B-636	Box, Fiberboard
PPP-F-320	Fiberboard, Corrugated and Solid Sheet Stock (Container Grade)
PPP-P-291	Paper, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain for Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper (for Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive, Filament Reinforced
NN-P-71	Pallet, Materials-Handling Wood
QQ-S-781	Strapping, Flat Steel

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43014	Boxes, Water-Resistant Paperboard Folding Set Up and Metal-Stayed

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized and Containerized Unit Loads 40 inch x 48 inch Pallets, Skids Runners, or Pallet-Type Base
MIL-STD-252	Visual & Mechanical Defects for Electronic Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

USAECOM

SC-DL-545358	Transmitter, Induction Compass T-611( )/ASN

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The Transmitter, Induction Compass T-611( )/ASN is a flux gate type earth magnetic azimuth field sensor which senses the direction of the horizontal component of the earth's magnetic field and generates an 800 Hz signal which is applied to a synchronizing mechanism control transformer or a differential synchro in the system indicator.

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When the indicator heading dial is in disagreement with the aircraft magnetic heading, a heading error signal is generated by the synchronizing mechanism. This information is used to precess and position the CN-998 Gyro in order to stabilize the system.

3.2 First Article. - When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3 and 6.2.d.

3.3 Construction. - The transmitter shall be constructed in accordance with equipment drawing and data list:

SC-DL-545358 Transmitter, Induction Compass T-611( )/ASN

3.4 Finish Protective (See 4.4). - The units shall be given protective finish in accordance with MIL-F-14072. This includes finishing of hardware such as screws, etc. and necessary touch-up after mounting. The final film on Type 1 surface shall be Film B in accordance with MIL-F-14072 and equipment drawings.

3.5 Cleaning. -

3.5.1 Parts. - Prior to the application of conformal coating on those parts affected and after fabrication, parts shall be cleaned in accordance with good commercial practice or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Part cleaning assembly processes shall minimize handling by bare hands to prevent contamination by fungus nutrients. Corrosive material shall be removed completely before the parts are mounted on chassis, panels and other parts.

3.5.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.6 Securing of Parts. -

3.6.1 General. - Brackets, lugs, inserts, bolts and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.6.2 Securing of parts by threaded fasteners. - Nylok or other screws and nuts with plastic devices and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

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3.7 Marking (See 4.4). -

3.7.1 General. - Marking shall conform to MIL-M-13231 and equipment drawings.

3.7.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification marking will be readily visible with minimum disassembly of the unit(s).

3.7.3 Serial number. - The following unit(s) shall be serial numbered:

Transmitter, Induction Compass T-611( )/ASN

3.8 Tropicalization of Materials. - Materials shall be treated as follows:

3.8.1 Treating materials. - Treating materials containing a mercury bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.8.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment or during use of the finished items when used for the purpose intended.

3.8.3 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposed to use.

3.9 Service Conditions (See 4.7). - The transmitter shall meet the service condition for Class I equipment of MIL-STD-810 unless otherwise specified.

3.9.1 Temperature-Altitude (See 4.7.1). - The transmitter shall meet the electrical requirement of 3.10 thru 3.10.3 before and after subjection to temperature-altitude test of Procedure I, Method 504, Table 504-I of MIL-STD-810 for Class I equipment. After returning to room temperature, the adjusting screws on the transmitter shall be operable.

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3.9.2 Humidity (See 4.7.2). - The transmitter shall meet the electrical requirement of 3.10 thru 3.10.3 before and after subjection to humidity test of Method 507, Procedure III of MIL-STD-810.

3.9.3 Vibration (See 4.7.3). - The transmitter shall meet the electrical requirement of 3.10 thru 3.10.3 before and after subjection to vibration test of Method 514.1, Curve B, Figure 514.1-2 of MIL-STD-810.

3.9.4 Vibration error. - When subjected to the vibration error test conditions of 4.7.4, the shift in heading output of the transmitter shall not exceed ± 1.0 degree.

3.9.5 Salt Fog (See 4.7.5). - The transmitter shall withstand exposure to salt-fog for 48 hours and shall show no evidence of corrosion as evidenced by any visible degradation or otherwise loosened finish.

3.9.6 Fungus (See 4.7.6). - The transmitter shall show no evidence of viable fungus after being subjected to 28 days of fungus incubation.

3.9.7 Dust (See 4.7.7). - The transmitter shall withstand exposure to sand and dust particles as encounted in desert areas.

3.10 Electrical. - Where required, performance of the transmitter shall be determined with Compensator, Magnetic Flux CN-405( )/ASN. The performance characteristics shall be in accordance with the following.

3.10.1 Heading error (See 4.8.1). - The accuracy of the transmitter shall be within ± 0.75 degrees.

3.10.2 Field strength variation (See 4.8.2). - When subjected to a horizontal field of 0.08 oersted and a vertical field of 0.54 oersted, the output of the transmitter shall not differ from that measured in a standard field by more than ± 0.75 degrees.

3.10.3 Tilt error (See 4.8.3). - When tilted 15 degrees in pitch (fore end up) the transmitter accuracy shall be within ± 1.5 degrees.

3.10.4 Transmitter output (See 4.8.4). - With the transmitter secondary circuit connected to the stator of the signal transformer, under normal excitation, and placed in a standard field strength, the maximum line-to-line output voltage of the transmitter shall be not less than 80 millivolts.

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3.10.5 Signal strength (See 4.8.5). - When connected to a standard signal transformer, the maximum 800 Hz voltage output of the transformer rotor shall not be less than 900 millivolts.

3.10.6 Output harmonics (See 4.8.6). - When connected to a standard signal transformer, the maximum 400 Hz output voltage of the transformer rotor shall not exceed 25 millivolts and the maximum 1200 Hz output voltage of the rotor shall not exceed 40 millivolts.

3.10.7 Voltage-frequency variation (See 4.8.7). - The shift in heading output from the transmitter shall not exceed ± 0.5 degrees when the input voltage is reduced to 22 volts and the frequency reduced to 360 Hz.

3.10.8 Neutral setting (See 4.8.8). - The compensator, when set to its neutral position and mounted on the transmitter, shall not cause a shift in the heading output of the transmitter of more than ± 1.0 degree compared to its output when the compensator is removed.

3.10.9 Compensator range (See 4.8.9). - The shift in heading output of the transmitter when the North-South or East-West adjustment of the compensator is set to maximum shall be from 8 to 13 degrees .

3.10.10 Cross-axis shift (See 4.8.10). - Rotation of the North-South or East-West adjustments of the compensator shall not cause a shift of the heading output of the transmitter of more than ± 1.5 degrees at a heading 90 degrees from the heading to which the compensation is applied.

3.11 Interchangeability (See 4.9). - Like units shall conform to Requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such item or of the units.

3.12 Workmanship (See 4.10). - The transmitter shall be manufactured and assembled in accordance with equipment drawings (See 3.3), Requirement 9 of MIL-STD-454 and the following paragraphs herein:

Cleaning	3.5
Securing of Parts	3.6
Tropicalization	3.8

3.13 Weight (See 4.11). - The weight of the transmitter shall not exceed 1.5 pounds.

4. QUALITY ASSURANCE

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4.1 Responsibility for Inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of Inspection. - The inspection requirements specified herein are classified as follows:

a. First Article Inspection (does not include preparation for delivery).

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality Conformance Inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.12 and 4.13)

4.3 First Article. - Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article units. - The contractor shall furnish several First Article units of the complete Transmitter, Induction Compass T-611( )/ASN as specified in Solicitation/Contract.

4.3.2 First Article inspection. - First Article inspection shall consist of the inspection specified in subsidiary documents covering the items listed in 4.4 and the inspections specified for Group A, B and C (See Tables I, II and III respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units; for Group C, two units for temperature-altitude and vibration, one unit for dust, one unit for salt-fog and one unit for fungus.

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4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirement specified in the contract or purchase order:

ITEM	WHERE REQUIRED
Finish Protective	3.4
Marking	3.7

4.5 Quality Conformance Inspection of Equipment before Preparation for Delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (See 6.4). Test equipment and facilities for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. - Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table I. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection, utilizing the procedures of MIL-STD-105, using the general inspection levels, and AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. - Group A inspection shall be performed in any order satisfactory to the Government.

4.5.2 Group B inspection. - This inspection, including sampling, shall conform to Table II and to the special procedures for small-sample inspection of Standard MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. The inspection level shall be S-3.

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TABLE I - GROUP A INSPECTION

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.12	4.10	2.5%	6.5%
Heading error	3.10.1	4.8.1		
Field strength variation	3.10.2	4.8.2	*1%	--
Tilt error	3.10.3	4.8.3	for the group	
Transmitter output	3.10.4	4.8.4		
Signal strength	3.10.5	4.8.5	*1%	
Output harmonics	3.10.6	4.8.6	for the group	
Voltage-frequency variation	3.10.7	4.8.7		
Neutral setting	3.10.8	4.8.8		
Compensator range	3.10.9	4.8.9	*1%	
Cross-axis shift	3.10.10	4.8.10	for the group	

*All electrical defects are considered major.

4.5.3 Group C inspection. - Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection specified in Table III. Samples shall be selected in accordance with 4.5.3.1.

TABLE II - GROUP B INSPECTION

Inspection	Req. Para.	Para.	AQL
Interchangeability	3.11	4.9	6.5%

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TABLE III - GROUP C INSPECTION

Inspection	Req. Para.	Test Para.
Subgroup 1		
Temperature-Altitude	3.9.1	4.7.1
Humidity	3.9.2	4.7.2
Vibration	3.9.3	4.7.3
Vibration Error	3.9.4	4.7.4
Subgroup 2		
Weight	3.13	4.11
Dust	3.9.7	4.7.7
Salt-fog	3.9.5	4.7.5
Subgroup 3		
Fungus	3.9.6	4.7.6

4.5.3.1 Sampling for Group C inspection.

4.5.3.1.1 Subgroup 1. - For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be randomly selected every month.

4.5.3.1.2 Subgroup 2. - For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be randomly selected every two months.

4.5.4 Reinspection of conforming Group B and Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or order provided the sample units are refurbished and resubjected to and pass Group A inspection.

4.6 Standard Test Conditions. - Unless otherwise specified, all tests shall be performed at ambient temperature, pressure and humidity under the following conditions:

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4.6.1 Altitude. - Unless otherwise specified, the transmitter shall be tested in its normal operating position.

4.6.2 Vibration. - Unless otherwise specified, the transmitter shall be subjected to sufficient vibration during test at a frequency of from 1500 to 2000 Hz, such that the transmitter will describe, in a vertical plane, a circle of not less than 0.002 inch nor more than 0.005 inch in diameter.

4.6.3 Supply voltage. - Unless otherwise specified, all tests shall be conducted with 23.5 ± 0.5 volts, single phase, 400 ± 5 Hz alternating current applied to the transmitter.

4.6.4 Standard field strength. - Unless otherwise specified, all tests shall be conducted with a horizontal field strength of 0.18 ± 0.02 oersted and a vertical field strength of 0.54 ± 0.02 oersted. Specified magnetic field strengths shall be obtained with Helmholtz coils or equivalent.

4.6.5 Standard signal transformer. - All tests shall be conducted with a standard signal transformer designed for this purpose. The signal transformer shall consist of a three-phase, three-wire stator surrounding a single-phase, two-wire rotor. The maximum input line-to-line impedance of the stator with the standard null meter as the rotor load shall be 133 / 81° ohms, ± 10 percent of the magnitude and angle, when measured at 1000 Hz with a standard laboratory instrument, such as a General Radio type 650-A impedance bridge, or equivalent. The rotor when tuned to 800 Hz shall have a minimum output voltage of 625 millivolts when the line-to-line signal input voltage to the stator is 90 ± 10 millivolts at 800 Hz.

4.6.5.1 The signal transformer shall be as free as possible from inherent transmission error, in order that errors caused by its use in the compass direction signal will be kept to a minimum.

4.6.5.2 The signal transformer shall have a suitable dial, mounted concentric with its rotor shaft, and a suitable pointer on the shaft for determining angular displacement from 0 to 360 degrees of the rotor to within ± 0.1 degree.

4.7 Service Conditions. - The service condition tests shall be conducted as specified herein. Upon completion of each service condition test, the equipments shall be subjected to and pass a visual inspection. No defects will be allowed.

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4.7.1 Temperature-Altitude. - The transmitter shall be subjected to the test as specified in Procedure I, Method 504, Table 504-I of MIL-STD-810 for Class I equipment. Electrical test shall be performed before and after test, to determine compliance with 3.9.1.

4.7.2 Humidity. - The transmitter shall be subjected to the humidity test as specified in Procedure III, Method 507 of MIL-STD-810. Electrical test shall be performed before and after humidity test to determine compliance with 3.9.2.

4.7.3 Vibration. - The transmitter shall be subjected to vibration test as specified in Method 514.1, Curve C, Figure 514.1-2 of MIL-STD-810. Electrical test shall be performed before and after vibration to determine compliance with 3.9.3.

4.7.4 Vibration error test. - The transmitter shall be set at the zero, or north heading, and the signal transformer rotor shall be set for the zero, or null, position. The transmitter shall then be vibrated at an amplitude of 0.003 to 0.005 inches circular diameter such that the plane of the vibrational circle shall be inclined at 45 degrees to the horizontal and shall contain the east-west axis. It shall be vibrated at frequencies of 200 Hz increments from 600 to 2600 Hz, inclusive. The vibration at each frequency increment shall be for a sufficient period of time to permit measurement of the null position shift. The test shall be repeated at two other headings not less than 100 degrees apart. The shift in null position shall not exceed $\pm$ 1 degree from the normal heading.

4.7.5 Salt-fog. - The transmitter shall be exposed to salt-fog test as specified in Method 509 of MIL-STD-810, for 48 hours. The requirement of 3.9.5 shall be met. Operational test is not required.

4.7.6 Fungus. - The transmitter shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. Operational test is not required.

4.7.7 Dust. - The unit(s) shall be subjected to the dust test specified in Method 510 of MIL-STD-810.

4.8 Electrical.

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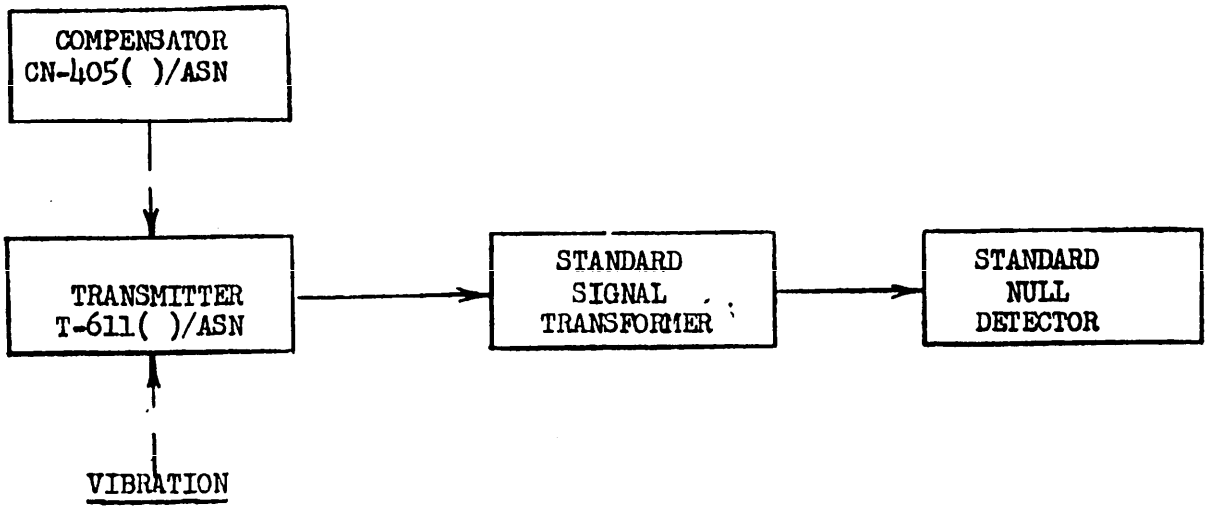
4.8.1 Heading error. - The transmitter shall be mounted on a turntable in its normal position, with standard field strength, and standard input voltage (excitation) applied. The transmitter secondary circuit shall be connected to the stator of a standard signal transformer whose rotor is connected to a standard null meter (See Figure 1). The transmitter and signal transformer shall be set at zero or north heading (rotor at null position). The transmitter shall then be turned in heading increments of 30 degrees, through 360 degrees. At each heading, the pointer of the signal transformer shall be turned correspondingly to its null position and the reading of the pointer on the dial taken. The difference between the pointer reading and the heading of the transmitter shall not exceed ± 0.75 degree after the index error has been subtracted. (NOTE: Index error is defined as the algebraic sum of all the readings obtained divided by the number of readings)

4.8.2 Field strength variation test. - The transmitter shall be set up with the signal transformer as specified in 4.8.1 and the transmitter shall be oriented to give zero index error. The horizontal field strength shall be decreased to 0.08 ± 0.005 oersted. The shift in null position of the signal transformer rotor shall not exceed ± 0.75 degrees after the index error has been subtracted.

4.8.3 Tilt error test. - The transmitter shall be set up with the signal transformer as specified in 4.8.1 except tilted 15 degrees in pitch (fore end up). It shall then be rotated smoothly about a vertical axis to the 90-degree heading in a period of 20 to 30 seconds. Similarly, the transmitter shall be rotated to and readings taken on the 180, 270 and 360 degree headings. The error between the signal transformer rotor and the transmitter shall not exceed ± 1.5 degrees after the index error has been subtracted. No vibration shall be applied to this test.

4.8.4 Transmitter output test. - With the transmitter connected as shown in Figure 1, and with normal excitation applied, the line-to-line voltages of the three wire output from the transmitter to the signal transformer stator shall be measured. For this test, the signal transformer rotor shall be tuned for maximum 800 Hz output, and shall be rotated for maximum line-to-line voltage. The transmitter output voltage shall be not less than 80 millivolts.

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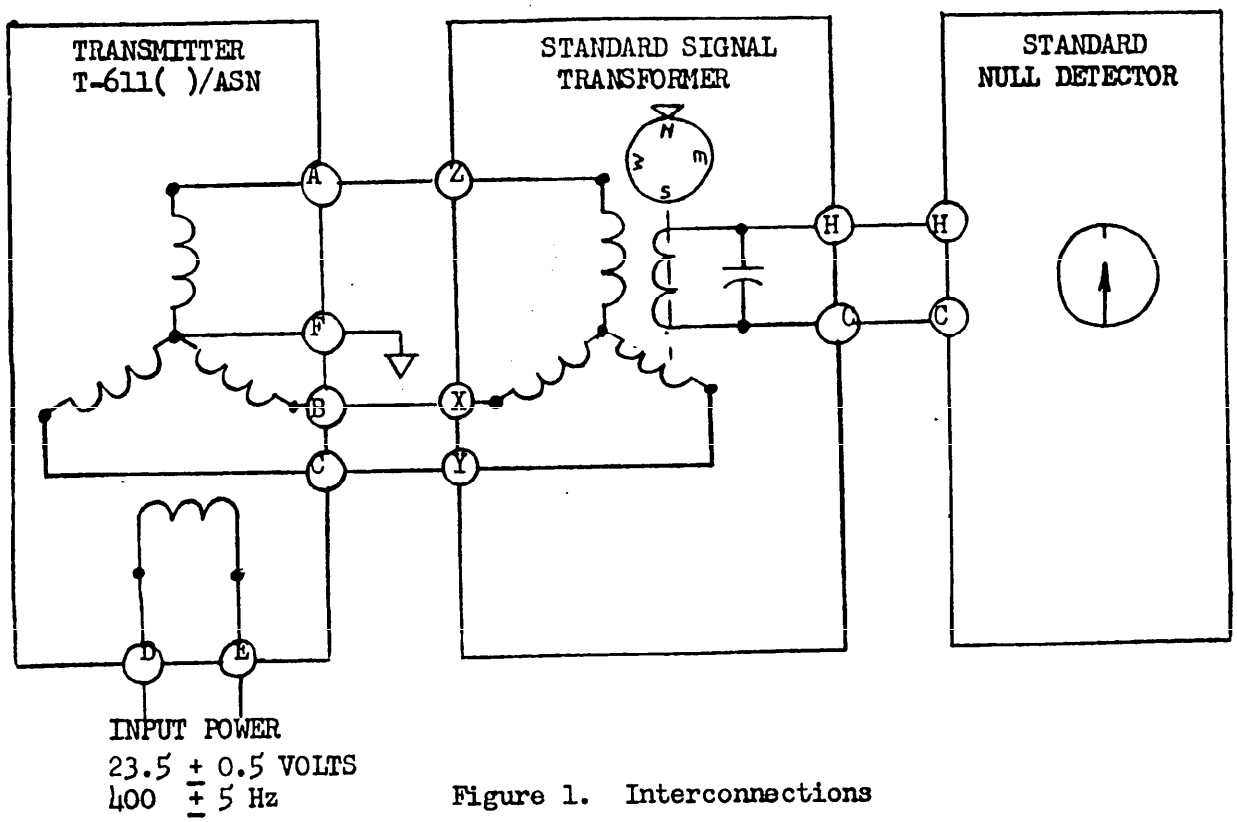


Figure 1. Interconnections

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4.8.5 Signal strength test. - The transmitter shall be mounted on a turntable in a normal position with a standard signal transformer. The transmitter and signal transformer rotor shall be set at zero or north heading. Then the heading of synchronization null shall be determined by rotating the transmitter. The signal transformer rotor heading shall be displaced 90 degrees from the transmitter heading and the maximum 800-Hz signal transformer rotor voltage recorded. This voltage shall be not less than 900 millivolts.

4.8.6 Output harmonic test. - The transmitter shall be mounted on a turntable in a normal position with a standard signal transformer. The transmitter shall be rotated for maximum 400 Hz signal transformer rotor voltage which shall not exceed 25 millivolts. The transformer shall then be rotated for maximum 1200 Hz signal transformer rotor voltage which shall not exceed 40 millivolts.

4.8.7 Voltage-frequency variation test. - The transmitter shall be set up as specified in 4.8.1, with the signal transformer on the zero degree heading. The supply voltage shall then be reduced to 22 volts and the frequency to 360 Hz. The shift in null position of the signal transformer rotor shall not exceed ± 0.5 degree. This test shall be repeated on two other headings not less than 100 degrees apart.

4.8.8 Neutral setting test. - The transmitter, without compensator, shall be set at zero heading specified in 4.8.1. The compensator, with both north-south and east-west screws set at neutral, shall then be placed in its normal position on the transmitter. The resulting shift of the null position of the signal transformer rotor shall not exceed ± 1 degree. This test shall be repeated with the transmitter set on the 90 degree heading.

4.8.9 Compensator range test. - The transmitter, without compensator, shall be set at the zero heading specified in 3.10.9. The compensator, with both north-south and east-west screws set at neutral shall be placed in its normal position on the transmitter. The north-south screw shall then be turned 180 degrees. The shift in null position of the rotor shall be between 8 and 13 degrees (plus or minus, depending on the direction of rotation of the adjustment screw). The same screw shall then be turned an additional 360 degrees in the same direction. The resulting shift in null position of the rotor shall be between 8 and 13 degrees. The test shall be repeated with the transmitter set on the 90 degree heading, using the east-west and leaving the north-south screw at neutral. The corresponding shift in null position shall be from 8 to 13 degrees on either side of neutral position.

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4.8.10 Cross-axis shift test. - The transmitter, without compensator, shall be set at the same heading specified in 3.10.10. The compensator, CN-405( )/ASN, shall then be placed in its normal position on the transmitter, with the north-south screw set at neutral and the east-west screw first set at 180 degrees clockwise, and than 180 degrees counter-clockwise, from neutral. The resulting shift in null positions of the rotor shall not exceed ± 1.5 degrees in either case. The test shall be repeated with the transmitter set on the 90 degree heading, with the east-west screw set at neutral and the north-south screw set 180 degrees both clockwise and counter-clockwise, from neutral position.

4.9 Inspection for Dimensional Interchangeability. - The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.10 Visual and Mechanical Inspection. - Inspection for visual and mechanical defects shall include, but not be limited to, inspection for the defects listed in the applicable sections of MIL-STD-252, and the equipment drawings, at all points of manufacture.

4.11 Weight. - The transmitter shall be weighed on a suitable balance scale with an accuracy of ± 5 percent of the weight to be measured.

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4.12 Quality Conformance Inspection of Preparation for Delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.13 Rough Handling Test (Preparation for Delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional tests shall be conducted to determine freedom from operational malfunction caused by the rough handling:

a.	Heading error	4.8.1
b.	Tilt error	4.8.3
c.	Signal strength	4.8.5

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging shall be Level A or C as specified.

5.1.1 Level A. -

5.1.1.1 Cleaning. - Transmitter Induction Compass T-611( )/ASN shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Transmitter Induction Compass T-611( )/ASN shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Each Transmitter, Induction Compass T-611( )/ASN shall be individually packaged Method III in accordance with MIL-P-116 as follows: cushion each unit by wrapping in paperboard conforming to PPP-P-291, Type I. Secure wrap with gummed paper tape conforming to PPP-T-45, Type III, Grade A. Place the cushioned unit within a close-fitting box conforming to MIL-B-43014, form 1, style II, Type A. Closure shall be in accordance with the appendix of the box specification.

5.1.2 Level C. - Transmitter, Induction Compass T-611( )/ASN shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be Level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

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5.2.1 Level A.

5.2.1.1 Consolidation. - A quantity of Transmitter, Induction Compass T-611( )/ASN, packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-636, type CF, class weather-resistant. Box closure shall be as specified in the appendix of the box specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantitites of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load. - A quantity of containers, packed as specified in 5.2.1.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147, except that the pallet shall be softwood conforming to NN-P-71, type IV, size 2. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.1.1 shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Consolidation. - A quantity of Induction Compass Transmitter T-611( )/ASN, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1.

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5.2.2.2 Palletized load. - A quantity of containers, packed as specified in 5.2.2.1, shall be palletized as specified in 5.2.1.2.

5.2.2.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.2.1 shall be reinforced by pressure-sensitive filament tape conforming to PPP-T-97, type IV, as specified in the appendix of the box specification. No further packing shall be required.

5.2.3 Level C. -

5.2.3.1 Consolidation. - A quantity of Induction Compass Transmitter T-611( )/ASN, packaged as specified in 5.1, shall be packed as specified in 5.2.1.1, except that the fiberboard boxes shall be class domestic.

5.2.3.2 Palletized load. - A quantity of containers, packed as specified in 5.2.3.1, shall be palletized as specified in 5.2.1.2, except that the fiberboard cap shall be class domestic.

5.2.3.3 Less than palletized load. - When quantities per destination are less than a pallet load, the containers packed as specified in 5.2.3.1 shall be used as the shipping container. No further packing shall be required.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended Use. - The Transmitter, Induction Compass T-611( )/ASN is intended to be used as part of the AN/ASN-43 Gyromagnetic Compass Set to provide earth magnetic azimuth field direction for correction of the directional gyroscope of the compass set.

6.2 Ordering Data. - Procurement documents should specify the following:

a. Title, number and data of this document and any amendment thereto.

b. Level of packaging and level of packing required for shipment (Level A, B or C).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

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d. Preproduction inspection (including order of testing, if required).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Submission of the statement of treatment referenced in 3.8.3 should be submitted as required by Solicitation/Contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: T-611 W/ASN. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this technical requirement and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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